The Barbero–Immirzi Field as a Hyperbolic Axion–Dilaton: a Fixed Kinetic Mixing $\lambda = -2\sqrt{2/3}$ and the Spiral Twist Hypothesis

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Abstract

Promoting the Barbero–Immirzi (BI) parameter to a field in $\mathcal{N} = 1$ supergravity is equivalent to adding a dilaton–axion chiral superfield with a fixed Kähler potential (Gates, Ketov & Yunes 2009). We point out that this Kähler potential has metric $K_{\Phi\bar{\Phi}} = 3/(\Phi+\bar{\Phi})^2$, i.e. a hyperbolic field space of curvature $-2/3$ (the $\alpha = 1$ attractor geometry), so the BI axion's kinetic term is multiplied by $e^{\lambda\chi}$ of the canonical dilaton with $\lambda = -2\sqrt{2/3} \simeq -1.633$ *fixed, with no free parameter*. This is the kinetic-mixing structure shown to produce an apparent ("illusory") phantom crossing in DESI-type analyses while obeying the null energy condition (Toomey et al. 2025), whose fit gives $\lambda = -1.3^{+1.3}_{-1.5}$. We are not aware of earlier work deriving this geometry, and hence λ, from the BI field. We then report what survives four consistency checks: (i) with matter on the supergravity frame, the dilaton has a general-relativity point at $\phi = -\frac{1}{2}$ that is a cosmological attractor, but Cassini forces the dilaton within $|\Delta\chi| < 0.0072\, M_{\rm Pl}$ of it today, which switches the mechanism off; scale (Weyl) invariance offers a known route to matter blindness; (ii) the simplest supergravity potential (constant superpotential) has a turning point exactly at $\phi = -\frac{1}{2}$ but negative energy; (iii) the same $\alpha = 1$ geometry as an inflaton gives $n_s = 0.967$ at $N = 60$, 2.4σ below ACT DR6; (iv) the dilaton cannot be the early-dark-energy field. A first background calculation with the decay constant fixed by the Nieh–Yan period ($F = (4/\pi)\sqrt{3/2}\, M_{\rm Pl}$) behaves as a thawing dilaton ($w_0 \simeq -0.84$, $w_a \simeq -0.24$); a literature cross-check of the illusion is not yet reproduced. The constant-superpotential result does not derive the positive periodic potential used in the Spiral Twist model. A separate QCD-anomaly scale check gives a natural $n = 1$ BI-axion mass near 2.9×10^{-12} eV, implying oscillations long before equality, not the fitted $n = 3$ EDE or a late-time quintessence field. We state the model and the tests that can falsify it.

1 Status in one paragraph

This note fixes a theoretical result and a hypothesis, with their checks, before the hypothesis is fitted to data. The algebra in §2 is exact and was verified symbolically. Everything after it is a *hypothesis*: some checks fail or are in tension, and they are reported as such. The companion analysis (doi:10.5281/zenodo.23867116 and its updates) finds that the minimal late-time Twist field's best preference in data (DES Y5 supernovae) is a calibration effect: it vanishes when the supernova sample is split at $z = 0.1$, and weakens on DES-Dovekie and Union3. The present note is therefore motivated by theory, not by a fit.

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2 The BI field's supergravity geometry

Gates, Ketov & Yunes [1] show that a dynamical BI parameter in $\mathcal{N} = 1$ supergravity is equivalent to a chiral superfield $\Phi = \phi + ia$ whose imaginary part couples to the (supersymmetric) Holst term and whose real part couples to the curvature, with (their eqs. 5.36–5.37)

$$\Omega = e^{\Phi+\bar{\Phi}}\,(\Phi + \bar{\Phi}), \qquad K = -3\ln(-\Omega/3), \qquad \phi < 0. \tag{1}$$

The linear piece $-3(\Phi + \bar{\Phi})$ of K drops out of the metric, leaving

$$K_{\Phi\bar{\Phi}} = \frac{3}{(\Phi + \bar{\Phi})^2}, \qquad \mathcal{L}_{\rm kin} = -\frac{3}{4\phi^2}\left[(\partial\phi)^2 + (\partial a)^2\right] \tag{2}$$

(reduced Planck units): the upper-half-plane metric with Gaussian curvature $-2/3$, i.e. the $\alpha = 1$ case of $K = -3\alpha\ln(\Phi+\bar{\Phi})$. With the canonical dilaton $|\phi| = e^{\sqrt{2/3}\,\chi}$ and canonical axion $\tilde{a} = \sqrt{3/2}\,a$,

$$\mathcal{L}_{\rm kin} = -\tfrac{1}{2}(\partial\chi)^2 - \tfrac{1}{2}\,e^{\lambda\chi}(\partial\tilde{a})^2, \qquad \boxed{\lambda = -2\sqrt{2/3} \simeq -1.633}. \tag{3}$$

The same number follows from the general α-attractor relation $\lambda = -2\sqrt{2/(3\alpha)}$ [3] at $\alpha = 1$. Toomey et al. [2] fit this kinetic coupling to Planck + DESI DR2 and find $\lambda = -1.3^{+1.3}_{-1.5}$ (prior $(-3, 0)$); the fixed value lies inside their 1σ range. This match was found after their fit and is a consistency, not a test. It becomes a test when λ is held fixed in a fit. The sign of λ depends on the orientation of the dilaton.

3 Matter coupling and the solar system

If matter couples to the frame in which Ω multiplies R ("reading A"), the Einstein-frame metric is $g_E = (-\Omega/3)g_J$ and matter feels $A^2 = -3/\Omega$, so

$$\alpha_{\rm DEF}(\chi) = \sqrt{2}\,\frac{d\ln A}{d\chi} = \frac{2}{\sqrt{3}}\left(|\phi| - \tfrac{1}{2}\right). \tag{4}$$

The coupling vanishes at $\phi = -\frac{1}{2}$, where Ω is stationary, and $d^2\ln A/d\chi^2 = +1/3$ there: a Damour–Nordtvedt attractor. Cassini ($|\gamma - 1| < 2.3 \times 10^{-5}$, $|\alpha| < 3.39 \times 10^{-3}$) requires $|\Delta\chi| < 0.0072\,M_{\rm Pl}$ today. A dilaton pinned this tightly cannot drive the kinetic-mixing effect, so in reading A the model reduces to a single axion today. This parallels the Brax–Martin no-go theorem for no-scale quintessence [4]. The BI field's own kinetic structure is discussed in [9, 10].

Reading B (matter blind to the dilaton) is natural if the theory is scale (Weyl) invariant: the dilaton is then the Goldstone mode of broken scale symmetry with only derivative couplings [5, 6], and Einstein–Cartan gravity admits this symmetry with non-propagating torsion [7, 8]. Whether reading B can coexist with the $\alpha = 1$ geometry above is open.

4 The simplest supergravity potential

With a constant superpotential W, $V = e^K(K^{\Phi\bar{\Phi}}|K_\Phi W|^2 - 3|W|^2)$ gives

$$V(\phi) = \frac{81}{2}\,W^2\,\frac{(-\phi - 1)\,e^{-6\phi}}{\phi^2}, \tag{5}$$

with extrema at $\phi = -\frac{1}{2}$ (a local maximum, exactly at the GR point) and $\phi = -\frac{2}{3}$ (a local minimum), both at negative energy; $V = 0$ at $\phi = -1$. The $\phi = -\frac{2}{3}$ minimum is only local: $V \to -\infty$ as $\phi \to 0^-$, so this potential has no stable global vacuum. For $\phi < -1$ it is positive, but the signed slope $s = -d\ln V/d\chi$ is negative in the stated convention $\chi = \sqrt{3/2}\ln|\phi|$ (for example, $s \simeq -8.16$ at $\phi = -1.5$ and $s \simeq -14.29$ at $\phi = -3$). Its magnitude alone is not the positive-s runaway criterion. This constant-W calculation is not an uplift and does not derive the exponential or periodic terms in the Spiral Twist model.

5 The Spiral Twist hypothesis

In the Einstein frame with reading B:

$$\mathcal{L} = -\tfrac{1}{2}(\partial\chi)^2 - \tfrac{1}{2}e^{\lambda\chi}(\partial\theta)^2 - V_0 e^{-\kappa\chi} - m^2F^2\left[1-\cos(\theta/F)\right], \tag{6}$$

with $\lambda = -2\sqrt{2/3}$ (above) and $F = (4/\pi)\sqrt{3/2}\,M_{\rm Pl} \simeq 1.56\,M_{\rm Pl}$, the value at which the Nieh–Yan index has period one for a Planck-length torsion scale (companion analysis). The angle θ is the BI (Twist) field and the radius χ grows as the spiral unwinds. Free: κ, m, θ_i; V_0 is set by closure.

Potential-origin gap. The exponential dilaton term and the positive periodic angle term in this Lagrangian are hypotheses, not consequences of the constant-superpotential calculation above. The microscopic torsion mechanism that fixes their signs, amplitudes, periodicity, and the release epoch remains to be derived.

QCD-anomaly scale check. The Nieh–Yan BI field has a derivative coupling to the fermion axial current; after the chiral rotation, Mercuri and Taveras derive its pion mixing and the fixed interaction scale $F_\varphi = 2/\sqrt{6\kappa} \simeq 2.0\times10^{18}\,\mathrm{GeV}$, together with $m_\varphi \simeq 3.0\times10^{-12}\,\mathrm{eV}$ [12]. For a low-energy comparison, write the leading periodic QCD-anomaly potential as

$$V_{\rm QCD}(\varphi) = \chi_{\rm QCD}(0)\left[1-\cos\left(\frac{\varphi}{F_\varphi}\right)\right], \qquad m_\varphi^2 = V''_{\rm QCD}(0) = \frac{\chi_{\rm QCD}(0)}{F_\varphi^2}. \tag{7}$$

Using $\chi_{\rm QCD}(0)^{1/4} = 75.6\,\mathrm{MeV}$ gives $\chi_{\rm QCD}(0) = 3.27\times10^{-5}\,\mathrm{GeV}^4$ and $m_\varphi = 2.86\times10^{-12}\,\mathrm{eV}$, consistent with the published mass at the accuracy of these inputs [13]. This is a curvature match to the leading periodic approximation, not a derivation of the exact temperature-dependent QCD potential or of the separate AEGIS field map.

The scale comparison is decisive for the cosmological role. With $H_{\rm eq} = 2.50\times10^{-28}\,\mathrm{eV}$, $m_\varphi/H_{\rm eq} = 1.14\times10^{16}$. Defining oscillation onset by $3H = m_\varphi$ and assuming radiation domination with $g_* = g_{*s} = 10.75$ gives $T_{\rm osc} \simeq 46\,\mathrm{MeV}$; entropy conservation then gives $a_{\rm eq}/a_{\rm osc} \simeq 8.1\times10^7$. For $m \gg H$, harmonic oscillations average to $w_\varphi \simeq 0$ and redshift as matter. A small-angle estimate that supplies 5% of the *total* density at equality requires $\varphi_i/F_\varphi \simeq 3.5\times10^{-5}$; it assumes the field is subdominant and the oscillations are already harmonic, and is not a Boltzmann fit. The susceptibility density is about twice the total background at the nominal onset for an order-one angle, so the radiation-dominated estimate must not be extrapolated to large initial angles. The calculation uses zero-temperature susceptibility, fixed relativistic degrees of freedom, and the published Nieh–Yan normalization. It therefore supplies neither the transient $n = 3$ equality-locked EDE potential nor the late-time thawing field used in the Spiral Twist fits.

First background pass. A two-field solver (validated: exact ΛCDM limit; single-field limit; null energy condition) with a CPL fit to $H(z)$ gives, over 80 settings, at best $w_0 \simeq -0.84$, $w_a \simeq -0.24$ (at $\kappa = 1$), the thawing relation $w_a \simeq -1.53(1+w_0)$; with $F = 1.56\,M_{\rm Pl}$ the angle carries too little energy for the mixing to matter. Using Toomey et al.'s effective dilaton equation of state, the illusion appears and is stronger with the fixed λ ($w_{\rm eff}(0) = -1.50$ against -1.09 at their λ), but in the opposite time direction to their posterior; their benchmark inputs are not fully specified, and this cross-check is **not yet reproduced**. These numbers are provisional.

6 Consistency checks (rules fixed before computing)

Check	Result	Verdict
Inflation from $\alpha = 1$ ($n_s = 1-2/N$, $r = 12/N^2$)	$n_s = 0.9667$ at $N = 60$ vs 0.974 ± 0.003	tension (2.4σ)
Dilaton relaxation as early dark energy	$\Omega_\chi(z{=}3500) \leq 0.025$; G changes $\times 13$ at BBN	too small
Solar system (reading A)	attractor at $\phi = -\frac{1}{2}$, band $0.0072\, M_{\rm Pl}$	passes only if pinned

7 How to falsify it

1. Fit the Spiral Twist with λ and F fixed (two dark-energy parameters, as in w_0–w_a) to CMB, DESI and the three supernova samples; it fails if it does no better than the thawing single field.

2. Derive reading B from a Weyl-invariant Einstein–Cartan action containing the BI field; if the dilaton necessarily couples to matter, the model is excluded by Cassini.

3. With reading B, the dilaton is a light field coupled only to the dark sector: parity-violating gravitational waves and CMB birefringence from the angle remain the direct probes (companion analysis).

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